CITS3402/CITS5507: Assignment 1

Feedback Report

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27 / 30
90.0%

Criteria	Proficient (100%)	Competent (66%)	Novice (33%)	Mark
Q1(a) Nonce insertion and file I/O	Correct implementation fit to specifications. Only bytes 16–31 and 45–52 are modified; solved files verify correctly. Code was not messy, well documented, and ran successfully.	Correct implementation fit to specifications, but code was hard to follow.	Code required minor modifications to run, or did not follow specifications correctly (e.g. modified bytes outside the nonce or student number field).	**0.83** / 1
Q1(b) Serial birthday attack	Correct, efficient serial implementation exploiting the birthday paradox. Code was not messy, well documented, and ran successfully.	Correct implementation exploiting the birthday paradox, but code was hard to follow or inefficient.	Implementation contained major correctness issues.	**2** / 2
Q2(a) Parallel birthday attack: correctness	Parallel implementation always finds a genuine, verified collision for every provided pair.	Parallel implementation finds correct collisions for most provided pairs.	Parallel implementation is unreliable, non-deterministic in a harmful way, or frequently produces incorrect results.	**2** / 2
Q2(b) Parallel birthday attack: use of OpenMP	Correct implementation with excellent, well-justified use of OpenMP for parallelism.	Correct implementation with basic OpenMP parallelism.	Implementation contained minor mistakes or did not make effective use of OpenMP (e.g. significant serial bottlenecks, poor scaling).	**2** / 3
Q3 Correct solution of sample files	All file pairs solved correctly and within 15 minutes.	One file pair unsolved.	Multiple unsolved file pairs.	**3** / 3
Q4(a) Description of the birthday-attack algorithm	Thorough analysis with excellent detail, including the collision-detection data structure and expected number of trials.	Good analysis, but lacking minor details.	Brief analysis, significantly lacking details.	**3** / 3
Q4(b) Description of parallelisation and synchronisation strategy	Thorough analysis with excellent detail, including how race conditions on the shared data structure were avoided or managed.	Good analysis, but lacking minor details.	Brief analysis, significantly lacking details.	**5** / 5

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Q4(c) Description of memory usage and trade-offs	Thorough analysis with excellent detail, including the memory footprint of the collision-detection structure and any speed/memory trade-offs made.	Good analysis, but lacking minor details.	Brief analysis, significantly lacking details.	**3** / 3
Q5 Performance metrics and analysis	Thorough analysis with excellent detail, including timings and speedup across thread counts and across pairs of varying difficulty.	Good analysis, but lacking minor details.	Brief analysis, significantly lacking details.	**4.17** / 5
Q6 Formatting and general presentation	Well presented, easy to follow.	Okay presentation, but minor issues.	Poorly presented, hard to read or follow.	**2** / 3
Total				**27 / 30**

Marker comments

q1a) Code is not cleaned up.
q2b) Shared table is serialised.
q5) Incomplete analysis.
q6) Report is not proof-read.